

MATTHEW TOPHAM

AI Systems Architect | Full-Stack Software Engineer

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SUMMARY

AI systems architect and full-stack developer building interactive AI products, persistent world systems, and multimodal storytelling pipelines. Active Marinara Engine developer with 100+ merged pull requests across React/TypeScript, Node.js, capability APIs, transactional state, AI generation, responsive UI, and production validation.

EXPERIENCE

Pasta Devs | Marinara Engine 2026 - Present

AI Systems Architect & Open-Source Developer | Remote

- Authored 100+ merged pull requests spanning React/TypeScript UI, Node.js services, shared contracts, downloadable capability APIs, transactional persistence, prompt orchestration, documentation, and automated tests.
- Architected hierarchical world-state infrastructure with nested locations, graph validation, bounded prompt context, AI drafting/expansion, lore grounding, snapshots/checkpoints, revision conflict handling, and responsive map authoring.
- Built reliable movement workflows using atomic owner turns, idempotent command IDs, server-owned route validation, retry/recovery semantics, and stream/persistence sanitization for hidden model directives.
- Expanded multimodal storytelling across Game and Roleplay with storyboard episode planning, image/video generation, provider-aware formatting, inline playback, responsive controls, and deterministic session replay.
- Extended the capability runtime with versioned host contracts for model routing, resources, persistence, transactions, history/checkpoints, readiness, retries, and package-owned client contributions.
- Validated changes with deterministic regressions, Playwright browser tests, lint/type/build gates, bundle budgets, container checks, and desktop/mobile QA.

SELECTED PROJECTS

School of the Ancients Creator & Full-Stack Developer

- Built a voice-driven Socratic learning platform across web and VR using React/Vite, Node.js, Supabase/Postgres, real-time voice AI, CI/CD, and Unity XR prototypes.

AI Radar Creator

- Created a Python and GitHub Actions pipeline that ingests RSS feeds, filters and summarizes AI research/news with LLMs, and publishes a concise daily brief.

TECHNICAL SKILLS

Languages:	TypeScript, JavaScript, Python, SQL, Bash
Frontend:	React, React Query, Vite, responsive web apps, PWAs, Playwright
Backend & Systems:	Node.js, REST APIs, WebSockets, Postgres, Supabase, transactions, idempotency, snapshots/checkpoints
AI:	LLM integrations, agent/runtime architecture, prompt composition, multimodal image/video pipelines, local AI
Delivery:	Git, GitHub Actions, Docker, Linux, pnpm, regression testing, browser QA

EDUCATION & CERTIFICATION

University of Colorado Boulder B.A. Computer Science	Expected May 2027
Front Range Community College A.S. Computer Science	2025
NVIDIA-Certified Associate AI Infrastructure & Operations	Aug 2025